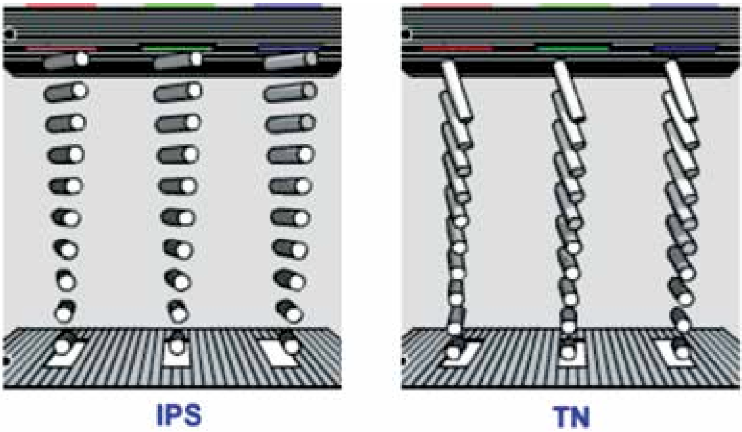

Half wave retardation of LC rods in IPS

But nowadays the LC configuration in IPS panels have gone through quite a few changes and this has improved the response time from 25 milliseconds to 10 milliseconds. Modern IPS panels consist of two crossed polarizers with LC rods at half-wave retardation which simply means that the LC rods, instead of being in a 90 degree position with respect to the edges of the polarizer are at an angle to it. This led to significant improvement in response times as the resting state of the LCs have been turned to 45 degrees.

Pros vs Cons

Pros

- Viewing angles or simply the angles at which you are going to watch your display is really good in IPS at around 178/178 degrees horizontal/vertical angle. In contrast, 95 per cent of TN panels have a viewing angle of at the most 160/160 degrees horizontal/vertical.
- Colour reproduction is fantastic on IPS panels as they are manufactured specifically for that. Its 8-bit colour reproduction per colour of RGB (Red Green Blue) ensures the reproduction of true 16.7 million colours. Some might argue that this type of colour reproduction can be achieved in TN panels too. That's a definite yes, it can be but via an emulated process called Dithering which doesn't exactly produce the same result.
- Digital signs or billboards are nowadays made of IPS panels replacing VA panels. The reason being that IPS panels are more durable and thus, can endure harsh weather conditions.